

WOUND MANGEMENT IN THE CHILDREN'S EMERGENCY

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WOUND MANAGEMENT IN THE CHILDREN'S EMERGENCY

1. GOALS OF WOUND MANAGEMENT

- Achieving Hemostasis
- Avoid Wound Infection
- Optimal Cosmesis

2. TYPES OF WOUND HEALING

TYPE OF HEALING	PROCESS	WOUND TYPE
Primary Intention	Wound edges are held together artificially by sutures/tissue glue. Minimal tissue loss & wound heals with minimal scarring.	Traumatic wounds Surgical wounds
Delayed Primary Intention	Wound is infected or at high risk of infection and requires intensive cleaning or debridement prior to primary closure 48 hours later.	Contaminated wounds Bite wounds*
Secondary Intention	Spontaneous wound healing through granulation, contraction, and epithelialization; Results in scar formation	Pressure ulcers Dehiscence wounds

* (most) Bite wounds should not undergo primary closure in the CE. Please refer to "Management of Animal Bites in the Children's Emergency" on the intranet and/or consult the CE senior.

3. WOUND EVALUATION

3.1 HISTORY

S	Signs & symptoms	Symptoms of infection (e.g., discharge, fever, warmth, pain) Symptoms of associated injuries (e.g., neurological deficit like loss of sensation)	
A	Allergies	Drug allergies	
M	Medication	Chronic medications (e.g., steroids can potentially affect wound healing, risk of bleeding with anti-coagulation therapy)	
P	Past medical hx	Tetanus vaccination status Comorbidities (e.g., diabetes mellitus, connective tissue disease)	
L	Last meal	Affects sedation timing	
E	Events to injury	Time of injury	Clean non-facial wounds – primary closure ≤ 18H from injury Clean facial/scalp wounds – primary closure ≤ 24H from injury Wounds > 12H from injury – not suitable for tissue glue.
		Mechanism of injury	Risk of wound infection (e.g., exposure to water, soil etc.) Risk of damage to surrounding structures High velocity/pressure wounds and/or puncture wounds might require surgical subspecialty review. Suspect NAI if there are unusual circumstances.
		Location of injury	Wounds that might require subspecialty closure – eyebrow, vermilion of lips, ears, tongue, genitalia, eyelids.

3.2 PHYSICAL EXAMINATION

SIZE OF LACERATION	Gaping wound > 5mm might require deep dermal sutures Lacerations > 5 cm are not suitable for tissue glue
SHAPE OF LACERATION	High skin tension when approximating skin flaps may further compromise vascular supply especially to the tip. Conventional interrupted stitches are not recommended.
FOREIGN BODY	Perform imaging to look for wound FB if clinical suspicion is high (e.g., radiographs, portable ultrasound)
SIGNS OF INFECTION	Pain, erythema, peri-wound skin changes
ASSOCIATED INJURIES	Neurovascular status, other MSK structures (tendons, bone)

4. WOUND ANESTHESIA AND PAIN MANAGEMENT

- Available options for anesthesia in the Children's Emergency are:
 - Local anesthesia (Topical, nerve blocks)
 - Intravenous Regional Anesthesia (Bier's Block)
 - Procedural sedation (IM/IV ketamine)

4.1 PSYCHOSOCIAL-EMOTIONAL FACTORS

- Pain is the unpleasant sensory and emotional experience associated with actual or potential tissue damage.
- Anxiety can exacerbate pain perceived by the patient.
- Reasons for anxiety include:
 - Anticipation of pain
 - Unfamiliar with procedure and process
 - Fear of needles
 - Unfamiliar environment
 - Transfer of anxiety from parents/NOK

4.1.1 ALLAYING ANXIETY

1. Age-appropriate distractions.
 - Younger patients – cartoon videos
 - Older patients – listening to music, video games, virtual reality
2. Patient education
 - For older patients who may benefit from mental preparation – explain to them step-by-step what to expect, use infographics and visual aids, involve parents to reinforce understanding.
 - For younger patients who will not be able to cope with “information overload” – explain briefly and in broad principles, avoid information overload, involve parents to allay anxiety.
3. See no needle – fear no needle.
 - Prepare T&S set away from the child
 - Only expose needles after draping the child and obstructing line of sight (for face & scalp lacerations) or after the child has been adequately distracted.
4. Allowing NOK to be present / go underneath the sterile sheet to distract child.
5. **NOT** allowing NOK to be present
 - Anxious parents/caregivers who transfer anxiety can be counterproductive.

4.2 LOCAL ANESTHESIA

4.2.1 LET GEL (LIDOCAINE, EPINEPHRINE, TETRACAINE)

TYPE OF LA	DOSE	ONSET	DURATION	REMARKS
LET gel (lidocaine, epinephrine, tetracaine)	1 – 3 ml	20 – 30 mins	2 – 3 h	▫ For lacerations < 5cm ▫ Most effective for wounds on the scalp or face. ▫ Less effective for extremities. ▫ Does not work when applied on intact skin.

METHODS FOR APPLYING LET GEL

1. Apply gel directly INTO the wound with a cotton-tipped applicator.
2. A cotton ball saturated with LET gel is placed into the wound and taped over.
3. How to use Tegaderm® to secure LET gel:



Secure lower end of Tegaderm®



Apply LET gel, ensuring that it enters the wound. Use cotton-tipped applicator if necessary.



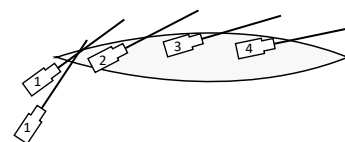
Close top half of Tegaderm®.
(Alternatively, a LET gel-soaked cotton ball can be placed before closing)

4.2.2 1% LIGNOCAINE FOR INFILTRATION

TYPE OF LA	DOSE	ONSET	DURATION	REMARKS
1% lignocaine	Up to 3 mg/kg	5 – 30 mins	2 h	Reduce pain during infiltration by: - Using buffered lidocaine (comes pre-prepared or dilute 10ml of 1% lignocaine with 1ml of 8.4% NaHCO₃; final volume = 11ml) - Warming the LA to body temperature - Using a high-gauge needle ($\geq 27G$) - Stabilizing the needle to avoid wobble - Slow injection of LA

INFILTRATING LOCAL ANESTHESIA FOR OPEN WOUNDS

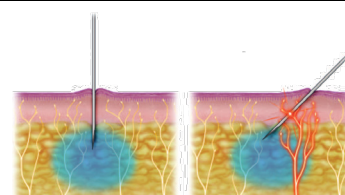
1. Insert needle into the subcutaneous layer from the wound edge.
2. Stabilize the syringe holding hand to reduce needle wobble.
3. Inject enough LA to cause a bleb and wait 30 seconds for effect of anesthesia.
4. Direct the needle left or right (directed to the right in diagram), aspirating before infiltrating LA, creating a bleb in the subcutaneous layer, before advancing the needle. ("blow before you go")
5. Once maximum distance of the needle is reached, withdraw needle, and insert needle ~ 1 needle length from the original site so that the patient will not feel the pain of the needle. (see diagram on right).



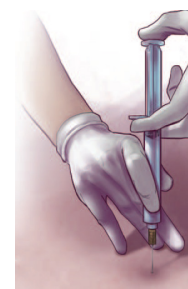
Infiltrate along the wound edge, making sure subsequent injections are within a previously anesthetized area.

HOW TO INFILTRATE LOCAL ANESTHESIA FOR DIGITAL BLOCKS

1. Apply EMLA cream 45 – 60 mins before the procedure to reduce pain over the injection site.
2. Distract the area of injection by pinching the skin or applying pressure adjacent to the injection site.
3. Enter perpendicular to the skin.
4. Stabilize the syringe holding hand to reduce needle wobble.
5. Inject LA into the subcutaneous layer instead of the intradermal area.
6. Upon entering the subcutaneous layer, immediately inject enough LA to create a subcutaneous wheal and then pause.
7. Only continue when the patient can no longer feel the pain of the needle (15 – 45 seconds).
8. Without moving the needle, slowly inject 0.5 – 1 ml of LA before tilting the needle.



Needle passes through less pain fibers if entered at 90° vs. 45° angle.



Anchoring syringe with non-dominant hand fingers to stabilize needle. Anchors also apply pressure that distracts perceived pain.

4.3 DIGITAL NERVE BLOCK

- Useful for procedures distal to the metacarpophalangeal joint

4.3.1 ANATOMY

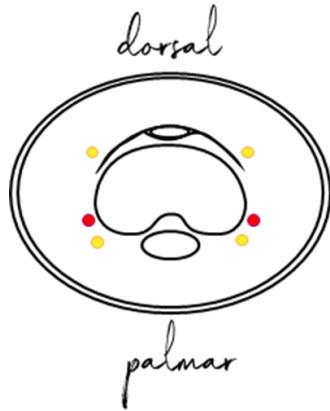


Figure 1 – Cross-section of finger. Digital nerves (yellow) at 2, 4, 8 and 10 o'clock positions.

- 4 nerves innervate each digit @ 2, 4, 8 and 10 o'clock positions (Figure 1.)
- Palmar nerves of the 2nd, 3rd, and 4th digits supply the palmar surface and dorsal surface over the distal phalanx. (Figure 2.)



Figure 2 – Sensory innervation of Palmar (red) & Dorsal nerves (yellow).

4.3.2 EQUIPMENT

- EMLA cream + Tegaderm®
- 1% lignocaine (3 mg/kg or 0.3 ml/kg)
- Syringe + 18G (Orange) bevel-less needle + 27G (Light Grey) needle for infiltration
- T&S set (refer [ANNEX A](#))

4.3.3 TECHNIQUES

- Apply EMLA cream for 45 minutes prior to injection and infiltration.

4.3.3.1 DORSAL APPROACH

- Patient's hand is positioned palms down.
- Insert needle perpendicular to the plane of the hand, just distal to the metacarpophalangeal joint and 3mm proximal to the most distal portion of the webspace, just adjacent to the proximal phalanx.
- Inject 0.5 mg/kg of LA into the subcutaneous space before advancing the needle to the plantar aspect and inject another 0.5 mg/kg.
- Repeat in the other webspace of the digit.

4.3.3.2 PALMAR APPROACH

- Landmark = palmar crease over the metacarpophalangeal joint.
- Insert needle into the subcutaneous space and angle needle towards any webspace and inject anesthetic (~ 1 – 1.5 mg/kg) – this would anesthetize both plantar and dorsal nerves on one side.
- Withdraw needle without exiting the skin and angle toward the contralateral webspace and inject anesthetic.
- Sufficient anesthetic is required to reach the dorsal aspect.

5. WOUND MANAGEMENT

5.1 ACHIEVING HEMOSTASIS

- Direct wound pressure
 - Well vascularized areas require at least 10 – 15 mins of direct wound pressure before hemostasis is achieved (e.g., pulp of fingertips)
- A proximal tourniquet can help with bleeding while suturing (e.g., pulp of fingertips)
- A deep dermal suture

5.2 WOUND CLEANSING & IRRIGATION

- Removes wound debris & reduces bacterial burden.
- Sustains a moist environment required for healing

5.2.1 CLEANSING SOLUTIONS

- Sterile Normal Saline
- 1% povidone-iodine (mixing 1 part of betadine to 9 parts of sterile saline) – for contaminated wounds
- Antiseptic solutions (e.g., chlorhexidine) should not be used directly in wounds as it can cause further tissue damage

5.2.2 CLEANSING PRESSURE

- Recommended pressure = 5 – 8 PSI (pressure per square inch)
 - Generated PSI influenced by:
 - ↑ needle size (lower gauge) = ↑ PSI
 - ↑ syringe size = ↓ PSI
- Recommended equipment to generate adequate PSI:
 - Contaminated wounds – 18G cannula + 20ml syringe
 - Other wounds – 20G cannula + 50ml syringe

5.2.3 CLEANSING VOLUME

- There is no clear evidence that favors irrigation with a minimum volume. We recommend approximately 50-100ml per centimeters of laceration, sufficient for proper cleansing and debris removal.¹

¹ European Wound Management Association (EWMA)

5.3 WOUND DEBRIDEMENT

- Devitalized tissue & foreign material should be removed from the wound.
- Reduces chance of wound healing well → negative cosmetic outcome.

5.4 METHODS FOR CLOSING THE WOUND

	ADVANTAGES	DISADVANTAGES
Tapes (Steri-Strip™)	▫ Fastest ▫ Painless ▫ No need for analgesia ▫ Low cost	▫ Least tensile strength ▫ Cannot get wet ▫ Not suitable for hairy surfaces
Tissue adhesives	▫ Fast ▫ Painless ▫ No need for analgesia / sedation	▫ Tensile strength < suture ▫ Potentially easily scratched off ▫ Highest risk for wound infection
Sutures	▫ Meticulous closure ▫ Great tensile strength	▫ Anesthesia required ▫ Requires removal (for non-reabsorbing sutures) ▫ Resource intensive

5.4.1 SUTURES

5.4.1.1 SUTURE SELECTION

- Choice of suture:
 - Smallest suture that adequately holds wound edges to achieve tension-free closure
 - Non-absorbable sutures for slow healing wounds.
 - Absorbable sutures for rapidly healing tissues.
 - If wound tension is high, smaller diameter sutures may injure tissue by cutting through them

	KNOT SECURITY	TENSILE STRENGTH	WOUND SECURITY	TISSUE REACTIVITY	SITE
Vicryl Rapide™	Good	Fair	7 – 10 days	Minimal	Face, scalp Under casts/ splints/ nailbed
Vicryl	Good	Good	30 days	Minimal	Dermal closure, mouth
Prolene	Least	Good	Best	Least	Anywhere. Workability fair.

- See “Timing of STO” below for suture type & sizing.

5.4.1.2 SUTURE TECHNIQUES

- Suture techniques that affect cosmesis:
 - Precise approximation of the wound edges.
 - Minimizing skin tension
 - Eversion of wound edges – minimize risk of scar depression secondary to tissue contraction during healing.

TYPE OF LACERATION	SUTURE TECHNIQUE
▫ Simple linear lacerations	▫ Interrupted stitches
▫ Gaping linear lacerations with: ▪ Adequate subcutaneous tissue ▪ Inadequate subcutaneous tissue ▪ Thin/friable skin	▪ Deep dermal suture + interrupted stitches ▪ Vertical mattress suture ▪ Horizontal mattress suture
▫ Lacerations with angled skin flaps	▫ Corner stitch (a.k.a. half-buried horizontal suture)
▫ Wound with edges that tend to invert (e.g., posterior neck wounds, on concave surfaces)	▫ Vertical mattress suture

- Please refer to **ANNEX B** for details on suture techniques.

5.4.1.3 TIMING OF STO (SUTURE-TO-OFF)²

WOUND LOCATION	TIMING OF REMOVAL	SUTURE TYPE	SUTURE SIZE
Face	3 – 5 days	Vicryl-Rapide™	5/0, 6/0
Scalp	7 – 10 days	Vicryl-Rapide™	3/0, 4/0, 5/0
Arms	7 – 10 days	Vicryl-Rapide™/Prolene	4/0, 5/0
Trunk	10 – 14 days	Prolene	3/0, 4/0
Legs	10 – 14 days	Prolene	3/0, 4/0
Hands or feet	10 – 14 days	Prolene	4/0, 5/0
Palms or soles	14 – 21 days	Prolene	4/0, 5/0

² Forsch RT. Essentials of skin laceration repair. Am Fam Physician. 2008;78(8): 945-951.

5.4.2 TISSUE ADHESIVES

5.4.2.1 GENERAL INFORMATION

- Tissue adhesives are cyanoacrylate monomers when exposed to moisture, form polymers that have a strong tissue bond (maximal bonding strength achieved within 2.5 minutes)

BUTYL-CYANOACRYLATE "LIQUIBAND"	OCTYL-CYANOACRYLATE "DERM+FLEX"
▫ Forms shorter chains ▫ Tighter & stronger bonds rapidly (15 secs) ▫ Bonds are brittle with increased risk of wound dehiscence	▫ Forms longer chains ▫ Weaker bonds, less rapid (30 secs) ▫ Less brittle

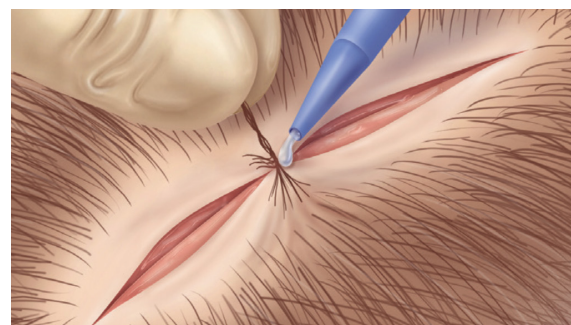
- To remove tissue glue, apply for 30 minutes to break polymer bonds:
 - Antibiotic ointment
 - Petroleum jelly
 - Nail polish remover
 - Acetone
- It is not harmful to the eye – it is used for corneal abrasion repairs
- Not for infected wounds or at high risk of:
 - Traps bacteria within the wound
 - Impermeable to water – does not allow for drainage of collections that build up

5.4.2.2 CONTRAINDICATIONS

- Wounds > 12 hours old
- Jagged or stellate lacerations or lacerations > 5cm
- Bites, punctures, or crush wounds
- Contaminated wounds
- Mucosal surfaces
- Axillae & perineum (high moisture areas)
- Hands, feet, and joints (unless kept immobilized and dry)

5.4.2.3 HAIR APPPOSITION TECHNIQUE (HAT)

1. Twist hair from opposite sides of the wound and rotate around the edges of the wound so that the wound edges oppose.
2. Maintain tension on the strands of hair on opposite ends of the laceration like how you would when securing sutures.
3. Apply a drop of tissue glue where the twists of the hair meet, taking care not to have adhesive run off into the wound.
4. Hold for at least 30 seconds to allow the adhesive to polymerize.
5. Repeat at intervals along the laceration (like interrupted sutures) till the laceration is closed.



5.4.2.4 WOUND CLOSURE WITH TISSUE GLUE

1. Wound should be irrigated, and surrounding skin should be treated with anti-septic.
2. Hemostasis should be achieved using adequate constant pressure on the wound.
3. Oppose the wound edges manually & evenly (ask for an extra pair of hands to help oppose the wound if required).
4. Prepare tissue glue by squeezing the vial/tube and stopping when a drop begins to form at the tip.
5. While the wound edges are opposed, apply tissue adhesive with a continuous brushing motion over the wound. Do not force applicator tip into the wound as this may result in a FB reaction with poor healing.
6. Hold for at least 30 seconds before releasing.
7. More layers should be applied in an oval pattern around the wound, with each layer encompassing a greater surface area on the skin.
8. Apply up to 3 layers (time taken to dry increases with each layer as there is less moisture available for polymer formation).

Tips & Tricks

- Prevent run off into the eye by covering it with a piece of gauze.
- Create a boundary with antibiotic ointment to limit run-off if positioning patient is not enough.
- Apply copious amounts of ophthalmic antibiotic ointment on eyelids that have been inadvertently glued shut. Do not attempt to pry open eyelids; lids will open after 1 – 2 days of his method.
- A 10-second grace period exists before the adhesive becomes too polymerized to wipe away. Use a dry gauze to wipe accidental run-offs away quickly.

5.4.2.5 WOUND CARE (FOR TISSUE ADHESIVES)

- Wound can be left uncovered as the polymer is water resistant.
- Patients may shower normally & pat the areas dry.
- Adhesive should spontaneously peel off after 5 – 10 days.
- Do not apply topical antibiotic ointments as that will affect the integrity of the polymer.
- A bandage to prevent picking at the wound or re-injury to the same location may be recommended for active children.
- (Long) baths are not recommended as prolonged exposure to water may loosen the top epithelial layer of skin and cause premature peeling with wound dehiscence.

6. LACERATION AFTERCARE

6.1 WOUND CARE ADVICE

- Keep wound covered for 48 hours, then remove dressing.
- After dressing is removed, apply antibiotic ointment twice a day
 - Facial wounds: 1% chlortetracycline ointment
 - Non-facial wounds: 3% tetracycline ointment
- Avoid sun exposure, apply sunblock to reduce hyperpigmentation.

6.2 TETANUS PROPHYLAXIS

Table from "Red Book®: 2015
Report of the Committee on
Infectious Diseases, 30th Edition"

HX OF ADSORBED TETANUS TOXOID	TYPE OF WOUND			
	CLEAN & MINOR WOUNDS		ALL OTHER WOUNDS ^a	
	DTaP, Tdap or Td ^b	TIG ^c	DTaP, Tdap or Td ^b	TIG ^c
< 3 DOSES OR UNKNOWN	YES	NO	YES	YES
≥ 3 DOSES	No, if < 10 years since last tetanus- containing vaccine	NO	No, if < 5 years since last tetanus- containing vaccine	No
	Yes, if ≥ 10 years since last tetanus- containing vaccine dose	NO	Yes, if ≥ 5 years since last tetanus- containing vaccine dose	No

Tdap indicates booster tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis vaccine; DTaP, diphtheria and tetanus toxoids and acellular pertussis vaccine; Td, adult-type diphtheria, and tetanus toxoids vaccine; TIG, Tetanus Immune Globulin (human).

^a Such as, but not limited to, wounds contaminated with dirt, feces, soil, and saliva; puncture wounds; fracture avulsions; and wounds resulting from missiles, crushing, burns and frostbite.

^b DTaP is used for children < 7 years. Tdap is preferred over Td for under-immunized children ≥ 7 years who have not received Tdap previously.

^c Intravenous immunoglobulin (IVIg) should be used when TIG is not available.

More frequent boosters are not needed and can accentuate adverse effects.

6.3 ROLE OF ANTIBIOTICS

- No role for empiric antibiotics
- Prophylactic antibiotics are indicated for:
 - Bite wounds (that break skin)
 - Immunocompromised patients
 - Contaminated wounds
 - Full thickness lacerations of the hands
- Antibiotic = PO Augmentin 22.5 mg/kg (max. 625 mg) TDS x 3 days
For patients with penicillin allergy, refer to "Management of Animal Bites In The Children's Emergency – ANNEX D"

6.4 DISPOSITION

- TCU Polyclinic or GP for STO (based on wound location)
- TCU Polyclinic or GP x 2 days for wound inspection & dressing (for contaminated wounds – concerns for infection based on mechanism of injury)

ANNEX A: EQUIPMENT LIST FOR TOILET & SUTURE (T&S)

GENERAL EQUIPMENT

1. T&S instrument set
2. Sterile gauze
3. Sterile cotton balls
4. Sterile gloves
5. Disposable Sterile Sheet
6. Chlorhexidine x 2 sachet
7. Jet Irrigation:
 - a. Contaminated wounds = 20 ml syringe + 18G cannula (Green)
 - b. Clean wounds = 50 ml syringe + 20G cannula (Pink)
8. Cleansing solution = 0.9% NaCl (20 ml) x 2 ampoules
9. Suture
10. Tegaderm®

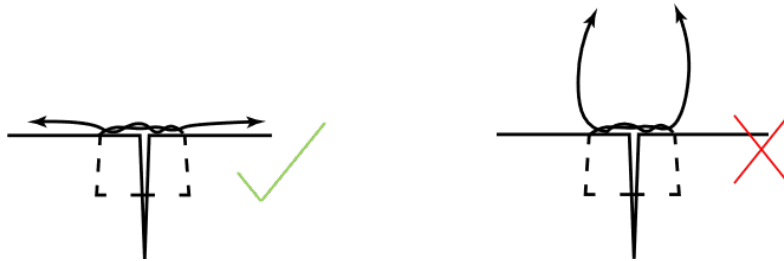
LOCAL ANESTHESIA (LA)

1. Buffered lignocaine or 1% lignocaine (up to 3 mg/kg)
2. 5ml syringe
3. 18G (Orange) bevel-less needle
4. 27G (Light Grey) needle for infiltration

ANNEX B: SUTURE TECHNIQUES

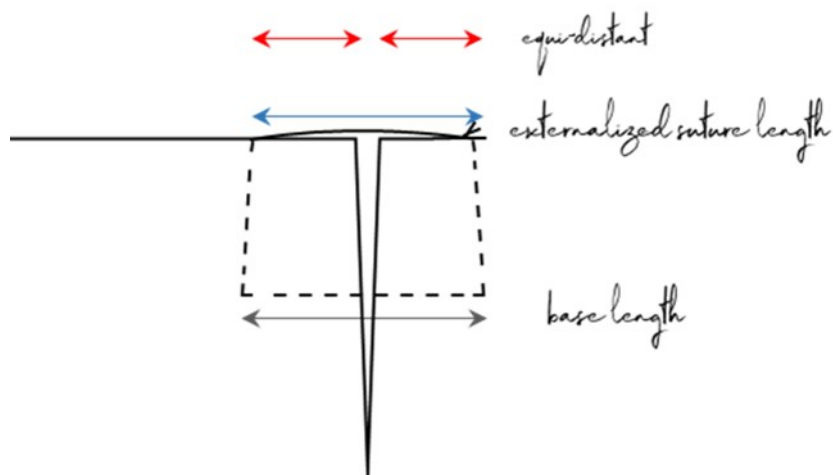
- Wound edges should be approximated closely. (Should not overlap)
- Wound edges should be everted. (Should not invert)
- There should be no tension across approximated skin edges.

THE SURGEON'S KNOT



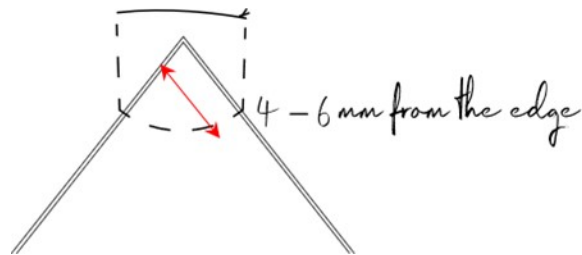
- The initial knot determines tension across the approximated skin edges. It holds in place only if suture ends remain flat to the skin surface. Lifting the suture on either end loosens the knot.
- When “squaring” (completing) the surgeon’s knot, the suture ends must be pulled horizontally while keeping the throw as flat as possible so that the anchoring knot does not unlock.
- Common mistakes that result in a loose knot:
 - Starting the knot too near the wound
 - Attempting to throw stitches with very little suture material left

SIMPLE INTERRUPTED SUTURES



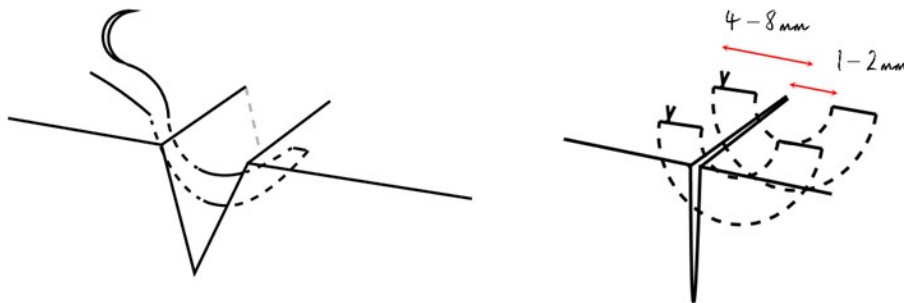
- Suture needle should penetrate the skin surface at a 90° angle.
- Base loop of the suture should be longer than the width of the externalized suture.
- Bites through the skin should be equidistant from the wound edge.

HALF-BURIED HORIZONTAL SUTURE



- Good for lacerations with skin flaps with poor vascular supply.
- The corner stitch distributes tension on the skin flap evenly – reducing the risk of vascular compromise especially to the tip.
- Suture placement, entry and exit must be equidistant in order for wound tension to be distributed equally and for wound edges to be aligned properly.
- The dotted line shows where suture is within the dermis.

VERTICAL MATTRESS SUTURE



- Seldom performed in the Children's Emergency.
- 2 main indications:
 - Everting wound edges – At body sites where wound edges tend to invert (e.g., posterior neck, wounds over a concave surface etc.)
 - Approximation of skin edges over a distance allowing for closure of wounds under high tension.
- “Far-Far-Near-Near” technique
 - Far-Far placement should be within 4 – 8 mm from wound edges with deep bites to the level of subcutaneous/deep dermis
 - Near-Near placement
 - Start on the opposite side of the wound where Deep-Deep was started
 - Within 1 – 2 mm from wound edge with shallow bites (1 – 2 mm deep)
 - Both ends of the suture thread should be tied on the same side of the wound.
- Risk of skin necrosis beneath the externalized loops of the vertical mattress sutures if knot is tied too tightly – this might result in a railroad scar if sutures are removed too late.

7. REFERENCES

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